Research and developments on the eastern Bering Sea walleye pollock stock assessment

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1 Summary

This document is a follow-up to the discussion paper presented to the Center for Independent Experts (CIE) review panel prepared in May 2025 (link here.
The review panel provided a set of recommendations to enhance the model structure, address key uncertainties, and improve transparency. This is Part 2 in which we show actions taken since the review and is intended inform the September 2025 NPFMC review. It focuses on migrating the operational model from ADMB to RTMB and begin to examine ways to evaluate the influence of Russian fishery.

Key outcomes since the CIE review • RTMB transition (Completed). The ADMB assessment was ported to RTMB and verified: • RTMB reproduces ADMB outputs for SSB and age-1 recruitment with negligible differences. • Enables random-walk selectivity/recruitment, marginal-likelihood error propagation, and RTMB flexibility to explore random effects. • Russian removals (In progress). Scenario modeling underway using Russian catch assumptions of ~20–45% of U.S. catch (per reviewer guidance and Sharov et al.). The goal is to bound potential impacts before exploring joint or explicitly transboundary models. • Process uncertainty and selectivity (Planned/Partially addressed). • Estimation of selectivity random-walk variances via marginal likelihood is now feasible in the RTMB "rpm" implementation (and in RCEATTLE/SPOCK), to be explored after the Russian scenarios. • Time-varying M remains a cautious exploration topic (informed by CEATTLE multi-species work). • Growth & weight-at-age (Deferred). Cohort and temperature effects on WAA acknowledged; staged for later development. • Data weighting / likelihoods (Deferred but available). Dirichlet–multinomial (DM) for composition fits (Laplace) is available (RCEATTLE, SPOCK, RTMB "rpm"); bootstrap-tuned effective sample sizes retained for now.

Bridging ADMB → RTMB (what changed, what matched) • Minimal code changes to ADMB logic to make it RTMB-compatible: • Fishery dome-shaped penalties applied to log-selectivities for ages >6. • "Deviation-about-a-mean" parameterizations re-expressed as level-parameters with mean-centered likelihoods (e.g., log_avgrec, log_avg_F, BTS selectivity slope). • Verification. Using ADMB MLEs in RTMB yields matching NLL components, near-zero gradients, and overlapping SSB/recruit series and asymptotic SDs. Differences are small and explained by the reparameterizations. • Demonstration of efficient MCMC application of ADNUTS package to the RTMB model.

Russian vs. U.S. assessments (EBS–WBS) • Signals match; magnitudes differ. Interannual patterns (especially age-1 recruits) are broadly synchronous; U.S. numbers-at-age are typically ~2–4× Russian values (peaking >5× in early 2010s). • Method/management contrasts. •

Russia: modern MSY rule with F_{target}=F_{MSY} and low B_{trigger} (5th percentile of SSB_{MSY}); recruitment treated as weakly stock-dependent → higher TACs when conditions are good, higher risk if recruitment dips. • U.S.: precautionary F_{ABC}<F_{MSY}, B_{35%} proxy, explicit SRR and uncertainty → lower volatility and risk. • Implication. Harmonized interpretation needs to account for different HCRs, survey systems, and model structures; scenario inclusion of Russian removals is the immediate step.

EBS vs. GOA comparison (context) • Scale and coupling. EBS abundances exceed GOA by roughly 5–15× across ages. Some weak–moderate coupling at early ages (shared environmental drivers), little coherence at older ages; populations are largely decoupled.

Technical notes added for transparency • Selectivity penalties. First-difference penalties over time and age are documented with zero-anchor conventions and RTMB R implementations. • Survey likelihood check. Lognormal objective (including constants, on log-scale) provided for parity with ADMB formulations.

Near-term priorities 1. Complete Russian-catch scenarios and quantify sensitivity of status and advice. 2. Switch on/compare marginal-likelihood estimation of selectivity process variances (RW penalties). 3. Pilot DM composition fits and contrast with bootstrap-tuned effective N. 4. Stage cohort/temperature WAA models and scoped exploration of time-varying M (informed by CEATTLE).

Bottom line. The platform migration is done and validated; the biggest outstanding uncertainty for 2025 advice is external removals (Russia). Process-error and likelihood improvements are queued behind that analysis to avoid confounding sources of change.

2 Response to 2025 CIE Review of EBS Walleye Pollock Assessment

This section outlines the response to the findings and recommendations of the 2025 Center for Independent Experts (CIE) review of the Eastern Bering Sea (EBS) walleye pollock stock assessment. The final peer-reviewed reports from the three reviewers are from Matt Cieri, Yan Jiao, and Anders Nielsen

One of the reviewers' primary concerns was the continued reliance on ADMB. This has now been resolved with the transition to RTMB, which enables use of random-walk selectivity and recruitment deviations, improved error propagation through marginal likelihoods and MCMC evaluations.

Another prominent theme was the absence of Russian removals from the assessment model. All three reviewers emphasized the need to explore the magnitude of this omission, both in terms of sensitivity runs and in better quantifying transboundary exchange. Work is in progress, with planned runs assuming Russian removals of 20–45% of U.S. catch based on (Japp and Sharov (2024)) and reviewer guidance. These scenarios will provide bounds on the possible scale of impacts before moving toward more collaborative or joint modeling approaches.

With respect to process uncertainty, selectivity, and growth, the reviewers identified several areas where additional flexibility could be valuable. Anders recommended estimating process variances for random-walk selectivity penalties using marginal likelihood, rather than fixing them. This aspect is possible within RCEATTLE and SPOCK currently, and is now available in the RTMB version of the pollock model (`rpm`). Yan noted that weight-at-age models could incorporate cohort and temperature effects, and also suggested cautious exploration of time-varying natural mortality, potentially informed by multispecies work such as CEATTLE. These issues have been partially addressed, and further exploration of process error variances will follow after the Russian catch work is integrated.

On data weighting and likelihood specification, Matt Cieri highlighted the potential advantages of a Dirichlet–multinomial (DM) framework estimated via Laplace approximation, as a more realistic representation of composition variance. For now, the bootstrap-based effective sample size approach has been retained, with DM implementation available in three different platforms (RCEATTLE, SPOCK, and the converted RTMB pollock model. We anticipate examining this as an option in the coming year.

Table 2.1: CIE 2025 recommendations and response summary

Theme	Lead Reviewer	Recommendation	Status
RTMB transition	All	Port model from ADMB to RTMB	Comple
Russian catch	All	Incorporate or test Russian removals	In progr
Selectivity complexity	Nielsen	Estimate process variances via marginal likelihood	Planned
Natural mortality	Jiao	Explore M variation and CEATTLE integration	Concept
Growth & WAA	Jiao	Model cohort and temperature effects on WAA	Deferred
Data weighting	Cieri	Test Dirichlet-multinomial likelihoods	Availab
Retrospective	Nielsen	Use Mohn's rho over shorter periods	Comple

A consolidated summary of reviewer recommendations and their current status is provided in Table 2.1.

Since the review, we adopted the ADMB operational model used in 2024 and ported it to RTMB. We verified that outputs match the between versions. Secondly, we have taken initial steps to address Russian catch uncertainty by comparing of assessments between regions.

3 Bridging ADMB to RTMB

The original ADMB code (first developed for application in 1996–Ianelli and Fournier (1998)) has evolved over time and has a number of unique features. However, ADMB has less flexibility to incorporate random effects and marginal likelihood estimation. As such, we adopted the set of features used for the 2024 EBS pollock assessment and fully ported those to RTMB. This transition enables several important enhancements, including the ability to estimate random-walk selectivity and recruitment deviations, improved error propagation through marginal likelihood, and greater transparency for code review and modifications.

We changed the ADMB code as little as possible to enable it to be compatible and work within the RTMB framework. One change was the use of an `if` statement in the penalty for dome-shaped selectivity in the fishery. In the original ADMB code, the penalty was invoked *if* the selectivity at one age was less than the selectivity at the previous adjacent younger age. The modification to this was simply to penalize log-selectivities of adjacent age groups older than age 6.

The ADMB code also has several cases where say y_i is a function of a parameter θ and a deviation ϵ_i such that $y_i = \theta + \epsilon_i$. In ADMB, the deviations are estimated as a vector of parameters with a constraint to have mean zero., In RTMB, it seems best to reparameterize them as a simple vector of parameters (say as θ_i) and then do the likelihood parts as deviations from the mean. To this end, we modified the following parameters which were parameterized as deviations from the mean (e.g., `log_avgrec` (average recruitment), `log_avg_F` (average fishing mortality), `log_slp_bts` (slope of the selectivity for the bottom trawl)) to be simply the vector of values. These changes resulted in changes in the pattern of spawning biomass and recruitment estimates among the 2024 ADMB model and the 2024 modified version of the ADMB model used for comparisons with the RTMB results (Figure 1).

Once these minor modifications were made, the RTMB code was able to replicate the results from feeding the MLE parameter estimates from ADMB into the RTMB code. This gave virtually the same estimates of spawning stock biomass (SSB) and age 1 recruits (Figure 2; Figure 3).

```
library(ebswp)
library(patchwork)
pm_orig <- read_rep(here::here("runs", "lastyr", "pm.rep")) # Read in the report file)
pm_mod  <- read_rep(here::here("runs", "rtmb", "pm.rep")) # Read in the report file)
sd_orig <- read_table(here::here("runs/lastyr", "pm.std")) |> filter(name=="pred_rec" )
```

```
sd_mod  <- read_table(here::here("runs/rtmb", "pm.std"))|> filter(name=="pred_rec" )
df <- rbind( data.frame( name  = "SSB",
  value = as.numeric(pm_orig$SSB[,2]),
  std.dev = as.numeric(pm_orig$SSB[,3]),
  model = "ADMB original" ) ,
  data.frame( name  = "SSB",
  value = as.numeric(pm_mod$SSB[,2]),
  std.dev = as.numeric(pm_mod$SSB[,3]),
  model = "ADMB Modified") )
df <- rbind(df, data.frame( name  = "Recruits",
    value = as.numeric(sd_orig$value),
    std.dev = as.numeric(sd_orig$std.dev),
    model = "ADMB original" ) ,
  data.frame( name  = "Recruits",
    value = as.numeric(sd_mod$value),
    std.dev = as.numeric(sd_mod$std.dev),
    model = "ADMB Modified") )
df$Year <- rep(1964:2024, 4)

p1 <- df |> filter(name=="SSB",Year>1977) |>
  ggplot(aes(x=Year, y=value, color=model)) + geom_point() + geom_line(aes(group=model)) +
  geom_ribbon(aes(ymin=value-2*std.dev, ymax=value+2*std.dev, fill=model), alpha=0.2, color=
  ggthemes::theme_few() + ylab("SSB (t)") + xlab("Year") +
  scale_y_continuous(labels = scales::comma, limits=c(0,NA)) +
  scale_x_continuous(breaks=seq(1965,2025,5))

p2 <- df |> filter(name=="Recruits", Year>1977) |>
  ggplot(aes(x=as.factor(Year), y=value, color=model)) +
  geom_point(size=2, position=position_dodge(width=0.8)) +
  geom_errorbar(aes(ymin=value-2*std.dev, ymax=value+2*std.dev),
              alpha=0.8, width=0.8, position=position_dodge(width=0.8)) +
  ggthemes::theme_few() +
  ylab("Age 1 Recruitment (thousands)") +
  xlab("Year") +
  scale_y_continuous(labels = scales::comma) +
  expand_limits(y = 0) +
  scale_x_discrete(breaks=seq(1980,2025,5))
p1/p2
```

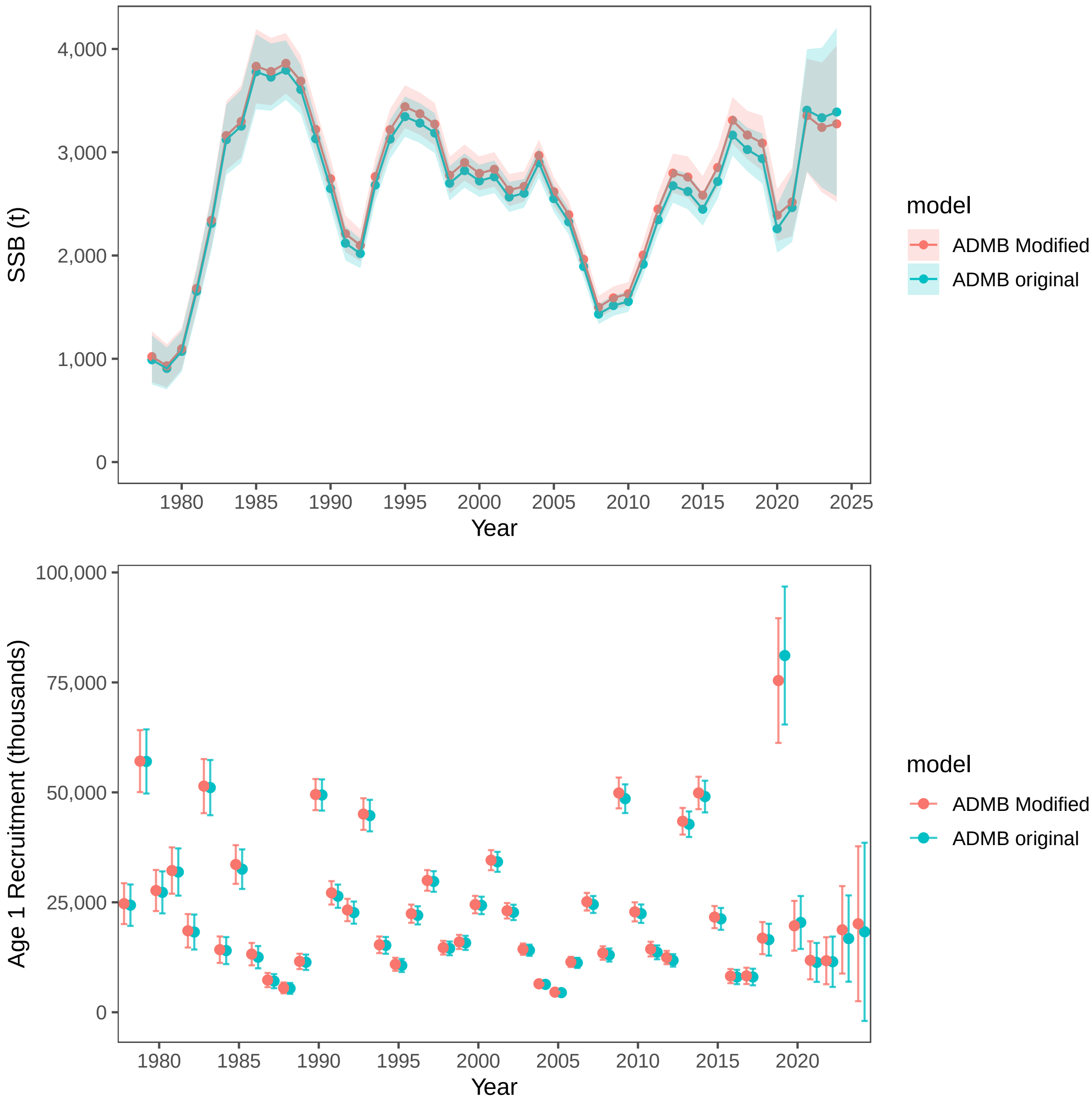


Figure 1: Differences in SSB and recruits between the base ADMB used in 2024 and the modifications for bridging with RTMB. Note that these are slight modifications of the ADMB configurations.

3.1 RTMB Model implementation

The RTMB model code is implemented in the `R/` directory, with the main entry point in `R/Rpm.R`. The initial steps to bridge ADMB to RTMB involve identifying key components and their behaviors. We created a table function that was helpful to evaluate which components of the model in RTMB tracked with the ADMB version. When the bridging aspect was acheieved there were de minimus differences between the two model versions (Table 3.1). Additionally, after some initial testing we were able to take the parameter estimates from the ADMB model and show that the gradients were very close to zero and the negative log-likelihood values matched (tbl-gradient).

3.2 RTMB implementation of MCMC integrations

A single run for a chain of 5,000 resulted in excellent mixing quality with the "poorest mixing" parameters demonstrating very good MCMC performance (Figure 4). The trace plots show the typical "fuzzy caterpillar" pattern indicating good mixing and the pairwise plots reveal appropriate exploration of parameter space. The chains appear to be sampling efficiently from the posterior distributions without any divergent transitions or other issues. The high effective sample size (ESS) values (ranging from ~900-1793) suggest that the chains are providing plenty of independent samples from the posterior distribution.

Using some of the additional features of teh `ADNUTS` R package shows that in relatively few commands, a useful suite of diagnostics can be created (Figure 5).

This figure provides comprehensive validation of the Bayesian MCMC results against frequentist asymptotic approximations for the EBS pollock stock assessment model, showing excellent overall performance across all diagnostics. Panel (a) demonstrates tight agreement between posterior standard deviations and asymptotic standard errors along the 1:1 line, validating that both approaches yield similar uncertainty estimates. Panel (b) shows well-behaved, approximately normal marginal posterior distributions for various log-scale parameters with good coverage by density curves, indicating proper sampling. Panel (c) presents the spawning stock biomass time series from 1978-2020, showing higher biomass in early years, decline through the 1980s-1990s, and lower variability in recent decades with appropriate uncertainty quantification throughout. Panel (d) displays pairwise plots for the 5 parameters with the greatest Bayesian-frequentist variance discrepancies, yet even these show reasonable correlations and good mixing in trace plots. Overall, this validation demonstrates that the RTMB Bayesian implementation performs excellently with close agreement to established frequentist methods while providing the additional benefits of full posterior distributions and proper uncertainty propagation.

The point of using MCMC's is that in the future previously assumed variances can be estimated more objectively.

Table 3.1: Comparison of the rtmb code (`RPM`) and the ADMB code results.

Opening /Users/jim/_mymods/noaa-afsc/EBS_pollock/runs/data/wtage2024.dat
[

Comparison of RTMB and ADMB Outputs
Tolerance = 1e-05] Comparison of RTMB and ADMB Outputs
Tolerance = 1e-05

Variable	Equal (tol)	Length	Max |Abs diff|	Max |% diff|	Correlation
N	TRUE	915	0.049435	0.000485	1.000000
Z	TRUE	915	0.000000	0.000159	1.000000
F	TRUE	915	0.000000	0.000477	1.000000
M	TRUE	915	0.000000	0.000000	1.000000
S	TRUE	915	0.000000	0.000122	1.000000
pred_catch	TRUE	61	0.004932	0.000380	1.000000
obs_catch	TRUE	61	0.005000	0.000458	1.000000
SSB	TRUE	61	0.004977	0.000416	1.000000
phizero	TRUE	1	0.000000	0.000189	NA
Bzero	TRUE	1	0.003401	0.000056	NA
steepness	TRUE	1	0.000000	0.000070	NA
pred_cpue	TRUE	12	0.004213	0.000120	1.000000
pred_avo	TRUE	18	0.000005	0.000413	1.000000
eb_bts	TRUE	42	0.048608	0.000455	1.000000
eb_ats	TRUE	19	0.004484	0.000223	1.000000
sel_fsh	TRUE	915	0.000005	0.000491	1.000000
sel_bts	TRUE	645	0.000000	0.000405	1.000000
sel_ats	TRUE	465	0.000005	0.000466	1.000000
sam_fsh	TRUE	60	0.000499	0.000134	1.000000
sam_bts	TRUE	42	0.000000	0.000000	1.000000
sam_ats	TRUE	19	0.000000	0.000000	1.000000
phat_fsh	TRUE	900	0.000000	0.000464	1.000000
phat_bts	TRUE	630	0.000000	0.000471	1.000000
phat_ats	TRUE	285	0.004822	0.000436	1.000000
age_like	TRUE	3	0.001044	0.000200	1.000000
age_like_offset	TRUE	3	0.039496	0.000326	1.000000
cat_like	TRUE	1	0.000002	0.000084	NA
bts_like	TRUE	1	0.000014	0.000041	NA
ats_like	TRUE	1	0.000000	0.000003	NA
ats_age1_like	TRUE	1	0.000025	0.000220	NA
cpue_like	TRUE	1	0.000002	0.000190	NA
avo_like	TRUE	1	0.000005	0.000055	NA
avgsel_like	TRUE	1	0.000000	0.000207	NA
wt_nll	TRUE	4	0.004116	0.000172	1.000000
wt_like	TRUE	1	0.004805	0.000076	NA
rec_like	TRUE	7	0.000040	0.000187	1.000000
Fpen_like	TRUE	1	0.000002	0.000026	NA
sel_like	TRUE	3	0.000021	0.000124	1.000000
sel_like_dev	TRUE	3	0.000465	0.000375	1.000000
Priors	TRUE	4	0.000025	0.000126	1.000000
tot_like	TRUE	1	0.004098	0.000052	NA

Table 3.2: Top 20 parameters with largest gradients (absolute value) from RTMB using ADMB parameter estimates.

outer mgc: 6.168831e-05 outer mgc: 6.168831e-05

Parameter	gradient
log_q_cpue	-6.168831e-05
log_F_devs	6.048117e-05
sel_devs_ats	-4.178819e-05
log_F_devs	-3.994676e-05
log_F_devs	-3.965254e-05
sel_devs_ats	-3.679200e-05
log_rec_devs	3.621454e-05
log_F_devs	3.543620e-05
log_F_devs	3.507003e-05
log_F_devs	-3.475311e-05
log_F_devs	-3.454961e-05
sel_devs_ats	-3.219652e-05
log_F_devs	-3.198028e-05
log_F_devs	2.904682e-05
log_F_devs	2.786280e-05
sel_a50_bts_dev	2.688229e-05
log_rec_devs	-2.644907e-05
log_F_devs	-2.617959e-05
log_F_devs	-2.492655e-05
log_F_devs	2.483619e-05

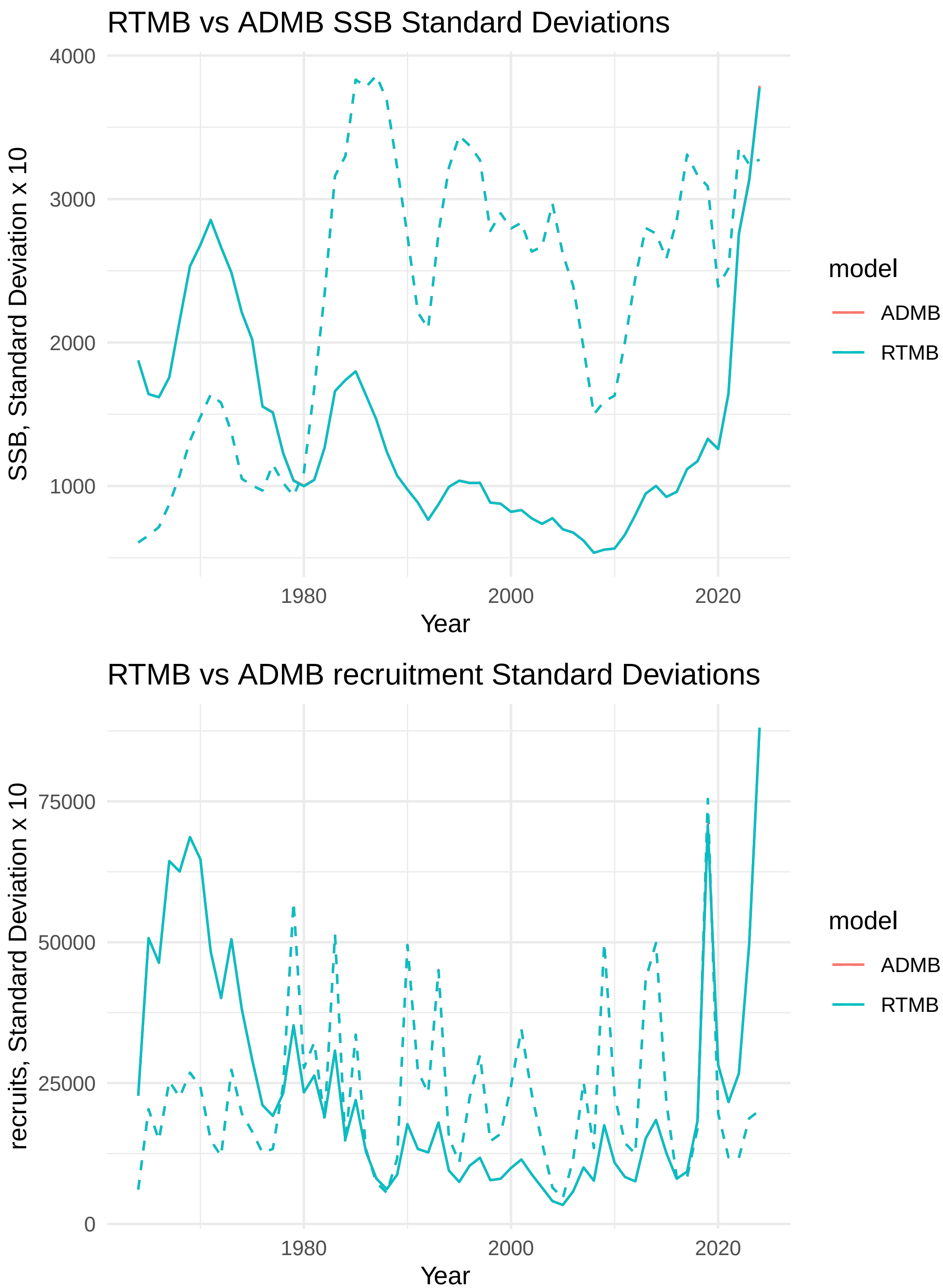


Figure 2: SSB and age-1 recruits from RTMB and ADMB, dashed lines are standard deviations (x10). Both ADMB and RTMB results are overlain.

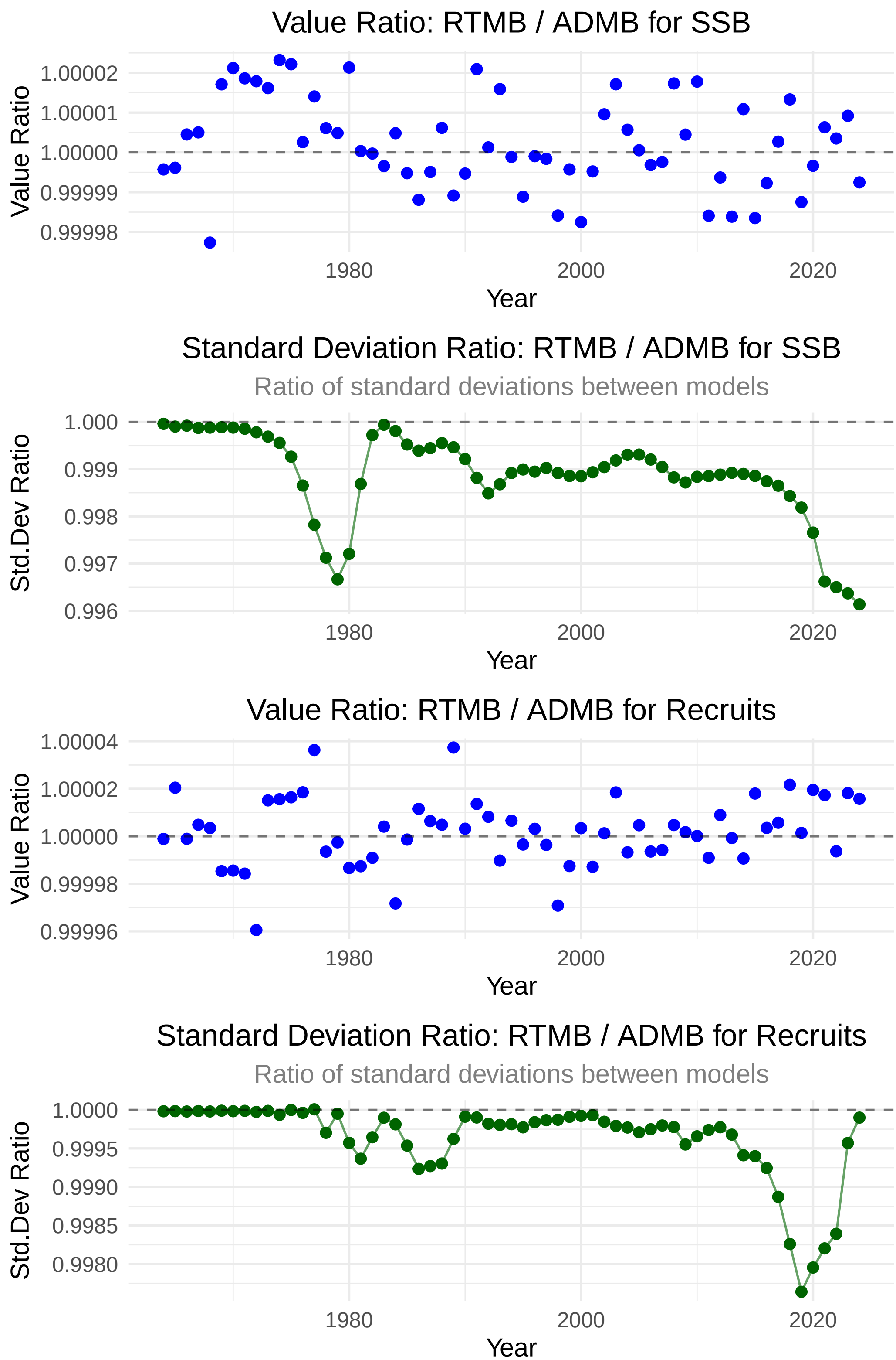


Figure 3: Relative differences between ADMB and RTMB estimates of SSB and age-1 recruits and asymptotic standard deviation approximations.

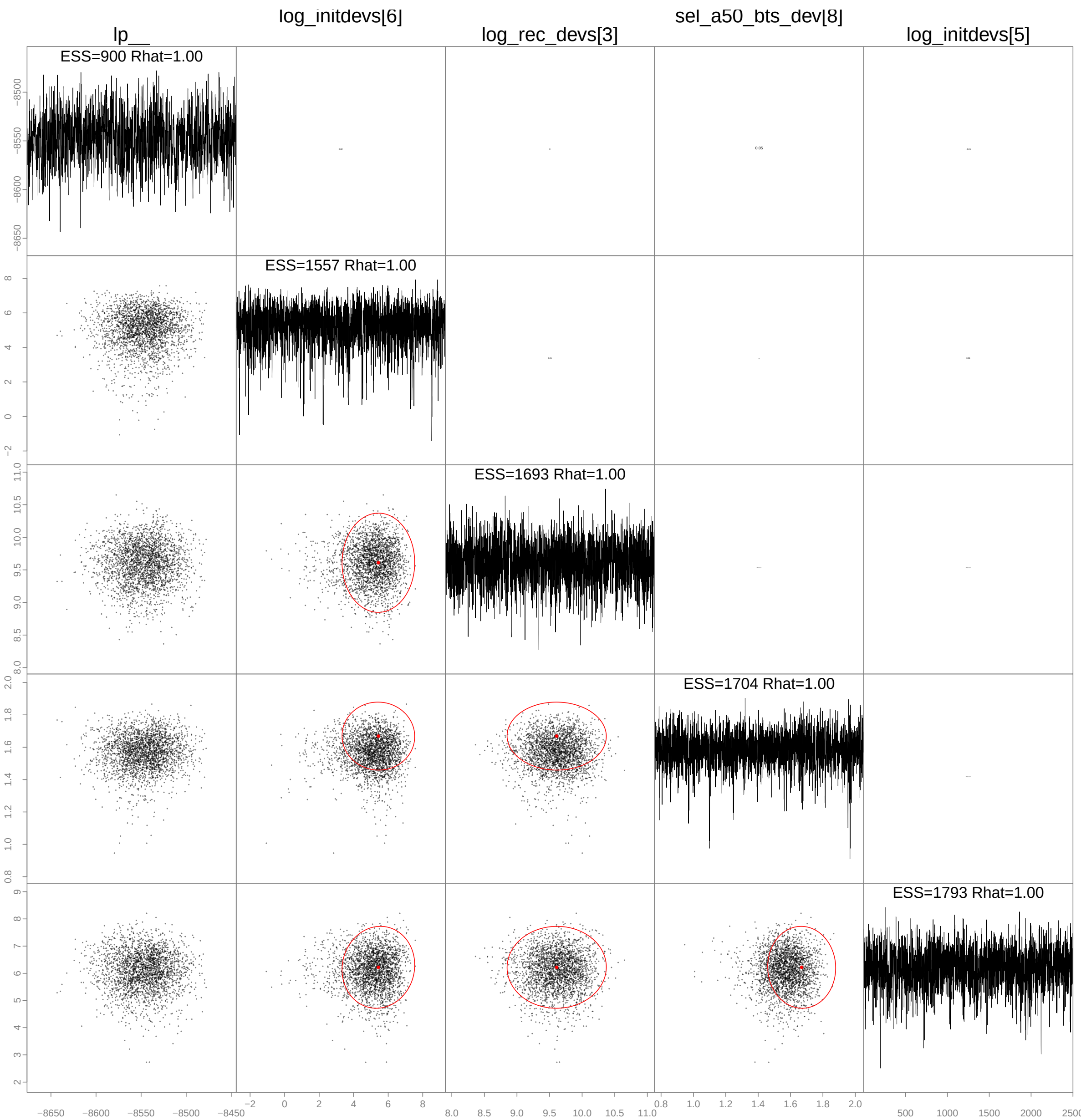
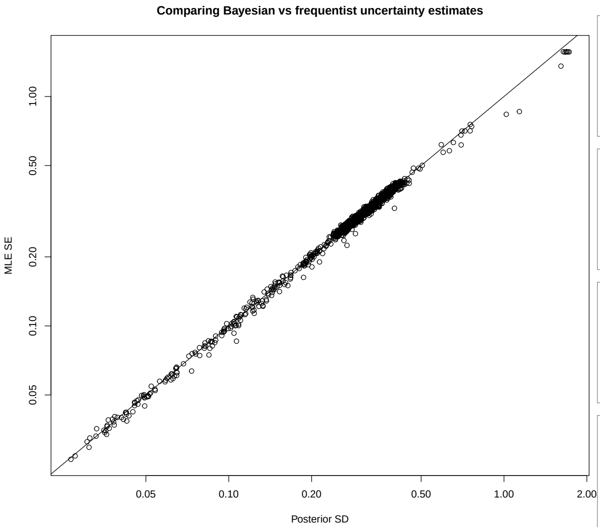
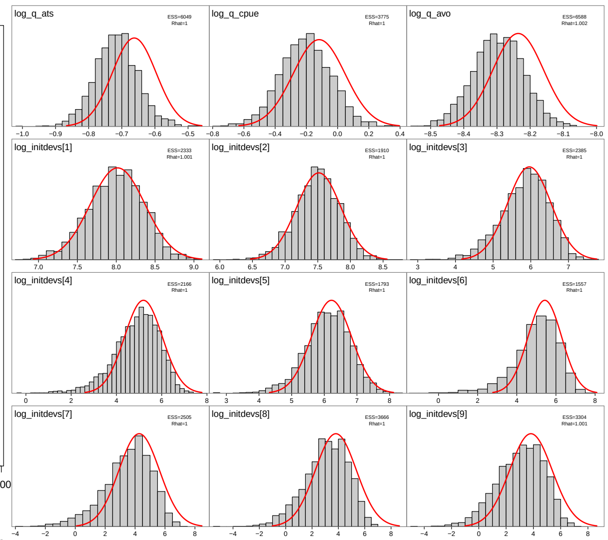


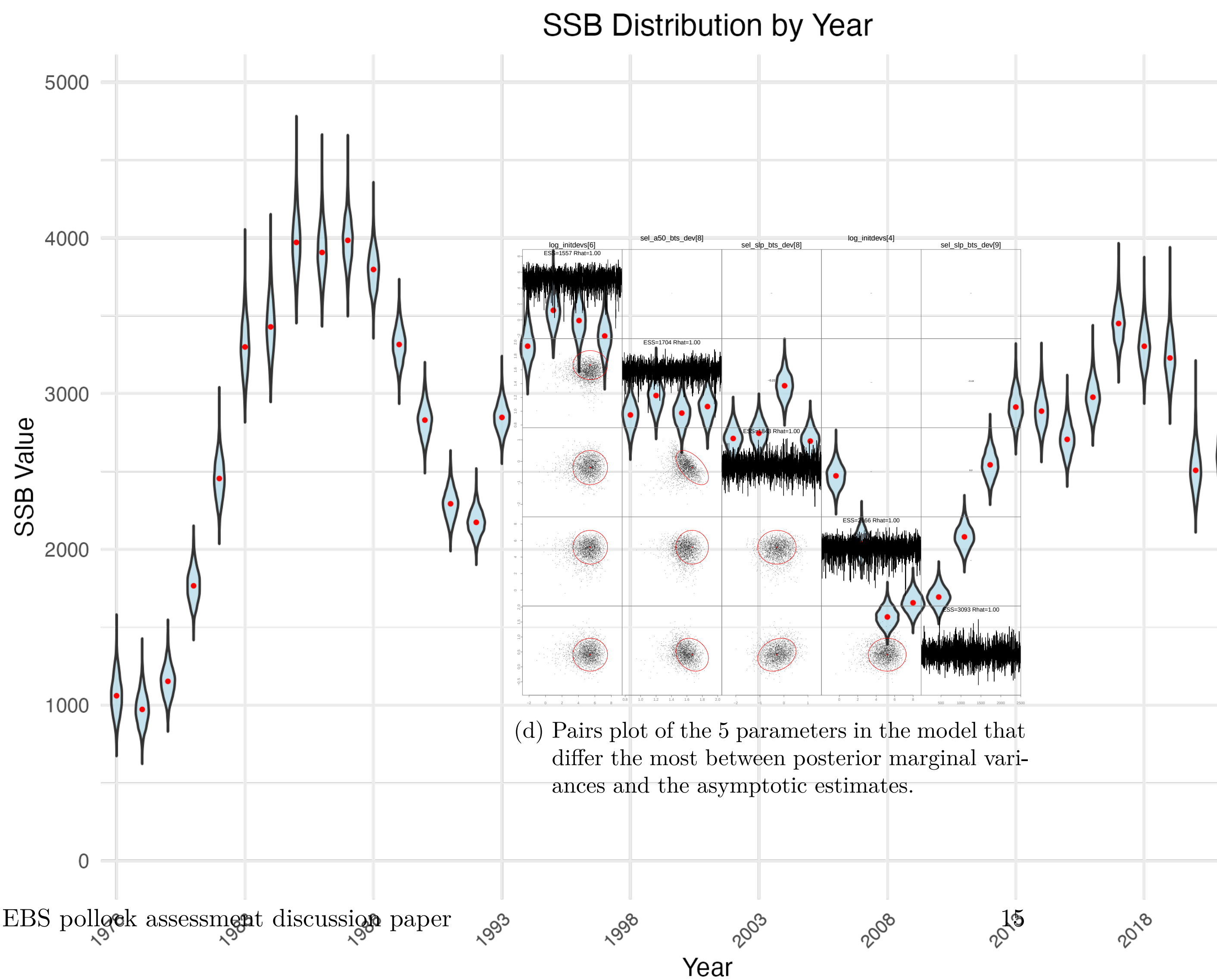
Figure 4: MCMC results from RTMB implementation of the EBS pollock model showing the slowest mixing parameters.



(a) Comparison of posterior marginal standard deviation with the asymptotic approximation.



(b) Posterior density distributions



(d) Pairs plot of the 5 parameters in the model that differ the most between posterior marginal variances and the asymptotic estimates.

(c) Distribution of spawning biomass.

Figure 5: MCMC diagnostics from RTMB implementation of the EBS pollock model.

4 Considerations for future assessments regarding neighboring Russian federation fisheries

As this year's assessment moves forward, particular attention is warranted on how Russian fisheries and assessments are considered alongside our own results. Pollock in the eastern and western Bering Sea are generally recognized as a single biological stock, with seasonal exchange across the U.S.–Russia boundary documented by acoustic studies. As noted in the CIE reviews, Russian removals are substantial with recent estimates representing on the order of 20–45% of the U.S. catch in some years.

While examined in past assessments, the Russian catch magnitude should be re-evaluated to examine how they might alter management advice. As a step in beginning to make a more detailed evaluation, we compared the Russian stock assessment results presented as part of their Marine Stewardship Council (MSC) review panel with our model results (through 2024). The following sections compare outcomes from the assessments presented to

4.1 Comparison of Russian and USA age-structured assessment results

4.1.1 Comparison of Russian WBS and U.S. EBS Pollock Assessments

Feature	Russian WBS Assessment	U.S. EBS Assessment
Primary model	TISVPA catch-at-age model (Vasilyev (2005))	ADMB statistical catch-at-age model (pollock model), routinely updated
Assessment frequency	Annual assessment; fishery-independent surveys ~ every 2 years	Annual assessment; multiple fishery-independent surveys each year (BTS, ATS, AVO)
Reference points (HCR)	Since 2022: **Ftarget = Fmsy (or proxy)**; **Btrigger = 5th percentile of SSBmsy distribution**	Precautionary: **FABC < Fmsy**, **B35%** proxy for SSBmsy; reference points set conservatively

Feature	Russian WBS Assessment	U.S. EBS Assessment
Management	"Modern MSY" approach, aiming for maximum sustainable harvest without exceeding safe stock status; allows ~15–30% higher TACs vs. precautionary approach	Precautionary approach emphasizes risk reduction; TAC constrained by Council harvest rules
Recruitment assumption	Recruitment assumed independent of stock size, fluctuating randomly within historical distribution	Stock–recruit relationship estimated (minor impact on recruitment estimates), with uncertainty reflected in projections
Recent stock status	2023: SSB well above SSBtarget and SSBmsy proxy; F < Ftarget and Flimit; stock increasing	2024: SSB above B35% and Bmsy proxy; F well below reference points; stock stable to increasing
Uncertainty notes	Higher terminal year uncertainty in TISVPA ; risk of overexploitation if stock declines under new HCR	Uncertainty routinely quantified (likelihood profiles, MCMC); precautionary buffers applied in catch advice
Governance & compliance	Managed by Federal Fisheries Agency (FFA) and CFMC; MSC certified since 2021; electronic logbook system in place	Managed under NPFMC/AFSC framework; extensive observer coverage; SAFE reports reviewed annually; MSC certified since 2005

The contrasts between the Russian WBS and U.S. EBS pollock assessments highlight both complementary perspectives and important methodological differences. Both systems conclude that pollock biomass is currently above target reference levels and that fishing mortality is below limits, reinforcing the perception of a healthy stock across the Bering Sea. However, the Russian shift to a "modern MSY" framework, with F_{target} set at F_{msy} and $B_{trigger}$ at the lower 5th percentile of SSBmsy, allows higher catch levels but introduces greater risk if recruitment declines. By contrast, the U.S. precautionary approach constrains harvest below Fmsy and maintains conservative biomass thresholds, prioritizing long-term risk reduction. These differences underscore the need for careful interpretation when comparing results across assessments: while the overall trends align, differences in model structure, reference points, and treatment of uncertainty can produce divergent management advice even under similar stock conditions.

4.1.2 Russian literature context: MSY vs. Precaution in WBS pollock

Bulatov and Vasilev (2023) argue for moving from a precautionary HCR to a modern *MSY-based rule* for WBS/Navarin and N. Okhotsk pollock. Their stochastic projections (using *TISVPA*) resample historical recruitment (assumed independent of SSB) and set $F_{target} \geq F_{MSY}$, with $MSY\ B_{trigger}$ taken as the 5th percentile of the SSB distribution under F_{MSY}. Reported values include:

- F_{MSY} 0.75 yr^{1} (Navarin, WBS) and 0.60 yr^{1} (N. Okhotsk).
- $B_{trigger}$ 470 kt (Navarin) and 1.77 Mt (N. Okhotsk).
- Expected catches 15–30% higher than under the precautionary rule (2002–2022 reconstructions), with large short-term gains in 2022 (e.g., Navarin TAC +~60%).

The implications for our EBS–WBS comparison are that the Russian MSY framing aligns with the recent WBS HCR updates (Ftarget=FMSY, low B-trigger), helping explain higher Russian TAC recommendations when recruitment is strong.

Datsky, Sheybak, and Antonov (2022) provide a comprehensive synthesis of pollock distribution, biology, stock status, and fishery dynamics across the Bering Sea. Drawing on >25 years of Russian survey and fishery data, they describe the **ecological plasticity** of pollock, its broad bathymetric range (20–600 m), and the role of **seasonal migrations** linking eastern and western Bering Sea populations.

Pollock in the Bering Sea are organized into three main regional groupings—eastern Bering Sea, Anadyr–Navarin, and western Bering Sea—that each maintain distinct spawning centers but intermingle during feeding migrations, especially in the northwest. The western stock experienced a marked decline in the late 1990s to early 2000s, but has gradually recovered since 2010, while biomass levels in the eastern and northwestern regions are now close to long-term averages. These dynamics are strongly shaped by environmental variability, including climate conditions, sea ice, and prey availability.

In Russian waters, pollock dominate the fishery, making up about 70% of total catches, almost entirely from midwater trawls. Recent landings range between 350,000 and 630,000 metric tons annually, with roughly 90% of harvests occurring between June and December. Importantly, the western and northwestern pollock populations are not isolated; they are continually reinforced by migrations from the eastern Bering Sea. This connectivity helps explain why U.S. and Russian assessments often capture similar recruitment and biomass patterns, even if absolute magnitudes differ, reflecting both shared basin-wide dynamics and methodological differences in survey coverage and assessment models.

Pollock dominate Russian Bering Sea fisheries, making up nearly 99% of total catches, with bycatch impacts on seabirds and marine mammals reported as very low (Artyukhin, Korobov, and Gluschenko (2023); Burdin (2021)). Recent analyses and reviews (Japp and Sharov (2023),

Table 4.2: Correlations between Russian and USA age classes, 1995-2024.

[

Correlations between Russian and USA Age Classes] Correlations between Russian and USA Age Classes

Age	Correlation
1	0.625
2	0.627
3	0.618
4	0.697
5	0.682
6	0.560
7	0.083
8	0.098
9	0.007
10	-0.500

Japp and Sharov (2024); Bulatov and Vasilev (2023); Datsky, Sheybak, and Antonov (2022)) emphasize the ecological and management context of these fisheries.

Comparisons between Russian and U.S. assessments indicate that U.S. numbers-at-age are consistently higher, with total abundance estimates averaging about 2.9 times greater than Russian estimates (median ratio 2.7). Despite this scale difference, recruitment and biomass trends display synchrony, suggesting that both assessments capture shared population dynamics across the basin while differing in magnitude due to survey intensity, model structure, and management frameworks.

The comparison of U.S. and Russian pollock assessments shows that U.S. estimates of total numbers-at-age are consistently higher than those from Russia, typically by a factor of 2–4, with the ratio peaking above 5 in the early 2010s (Figure 6, bottom-left). Both assessments show similar interannual patterns, particularly for age-1 recruits (top-right), suggesting some shared signals of strong year classes. Correlation analysis by age class (bottom-right; Table 4.2) confirms this alignment, with moderate to strong positive correlations across ages 1–6. At older ages, correlations weaken and even turn negative (age 10), reflecting differences in how the two assessments treat age structure and fishery/survey data inputs. Overall, while the magnitude of estimates differs substantially between the two countries, the consistent synchrony in recruitment trends indicates that both assessments are capturing the same broad population dynamics, albeit with systematic differences in scale.

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=== DETAILED STATISTICS ===
```

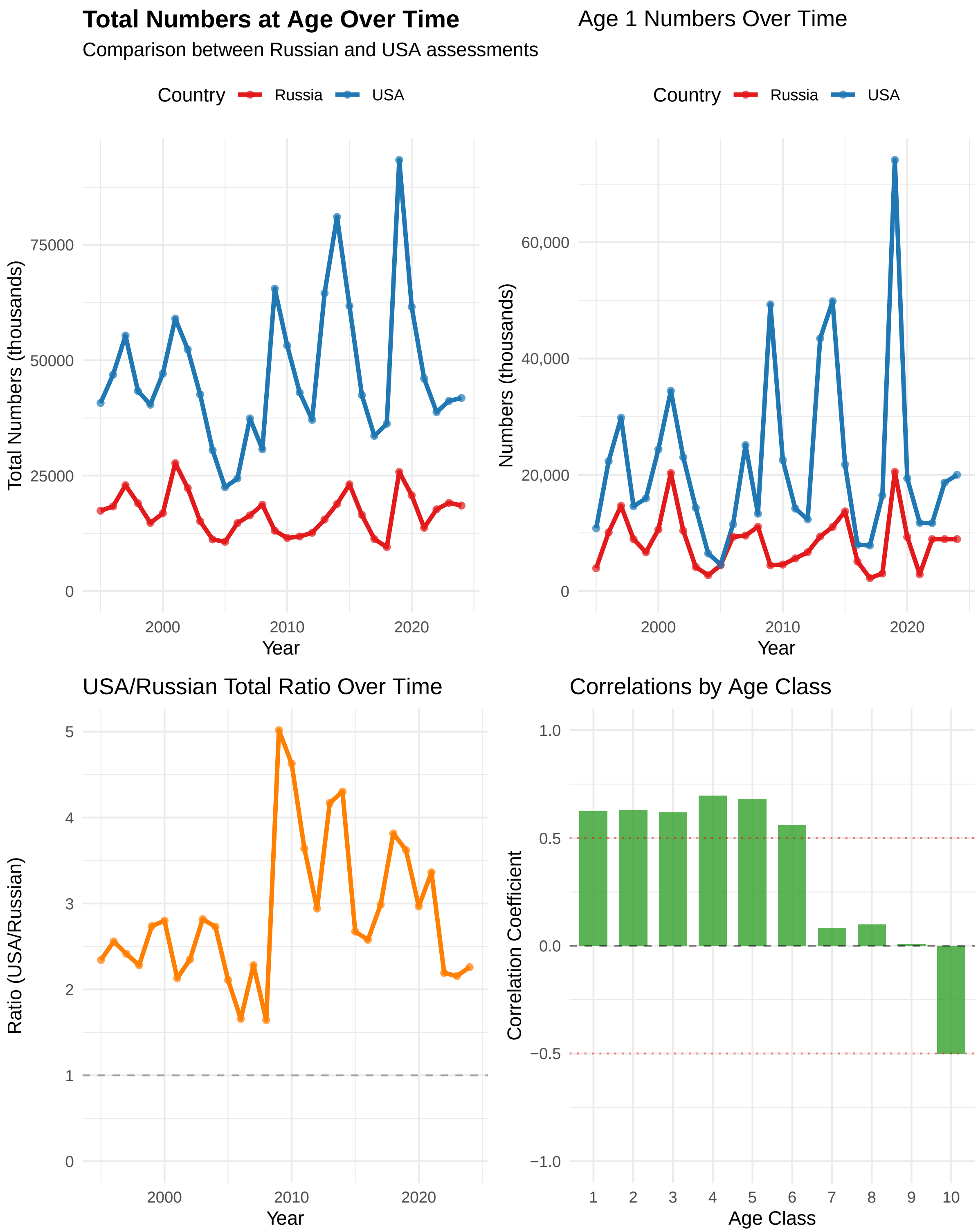


Figure 6: Various comparisons between Russian and USA age-structured survey results.

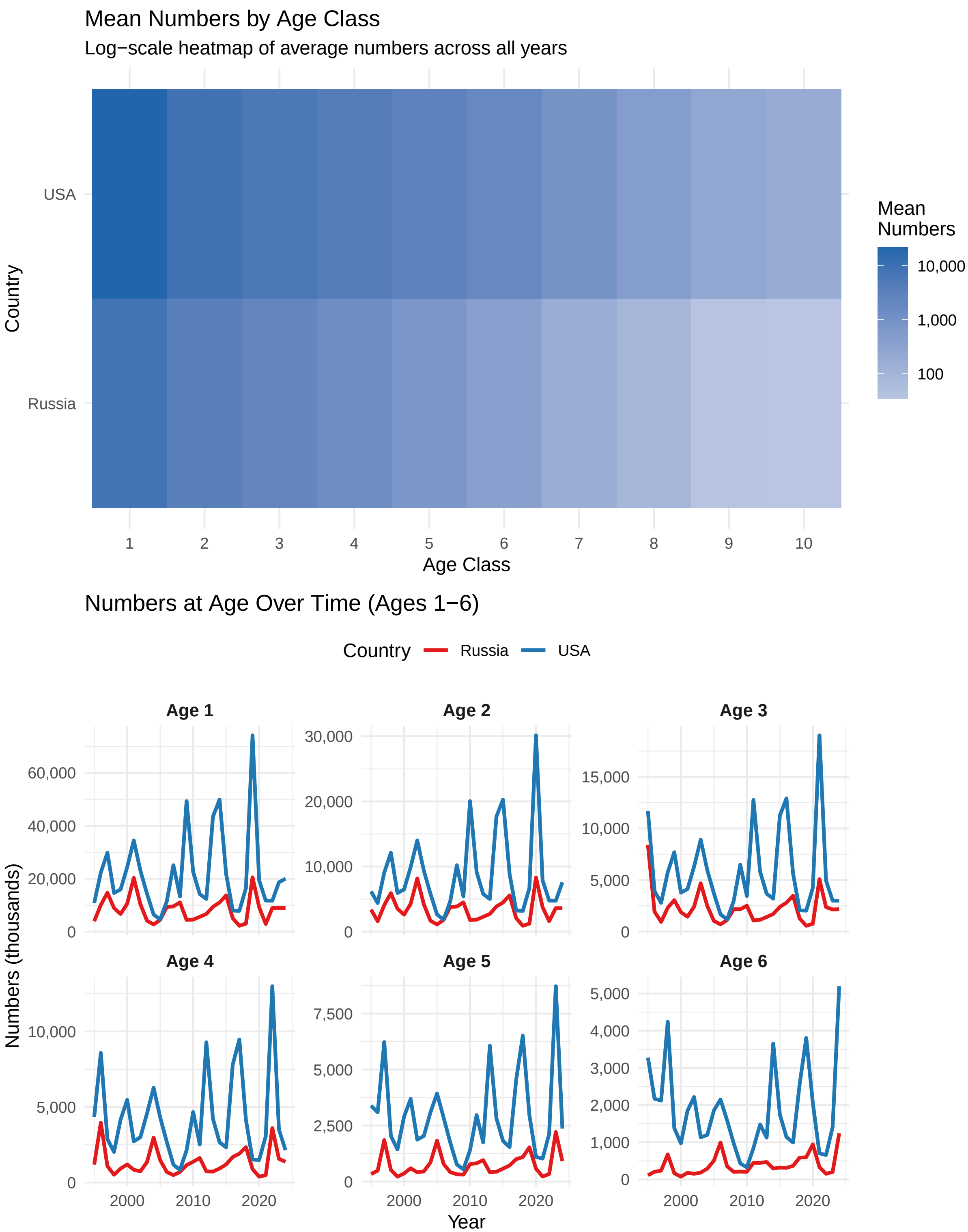


Figure 7: Various comparisons between Russian and USA age-structured survey results.

```
Summary statistics by country and age:

# A tibble: 20 x 8
   Country   Age   Mean  Median      SD    CV    Min     Max
   <chr>   <dbl>  <dbl>   <dbl>   <dbl> <dbl>  <dbl>   <dbl>
 1 Russia      1  8391.   8925    4647. 0.554 2213.   20509.
 2 Russia      2  3400.   3562.   1860  0.547  900.    8308.
 3 Russia      3  2254.   2154.   1574. 0.699  565.    8406.
 4 Russia      4  1338.   1169.    875  0.654  387.    3966.
 5 Russia      5   743     575.    511. 0.688  216.    2214.
 6 Russia      6   382.    302.    278  0.728   74.5   1242.
 7 Russia      7   168.    138.    127. 0.757   15.3    550.
 8 Russia      8    89.3    65.2    81.9 0.917   0.8    367.
 9 Russia      9    38.7    31.2    37.3 0.964   0.4    178.
10 Russia     10    35.1    16.8    45.4 1.29     0.2    164.
11 USA         1 21725.  17539.  15273. 0.703 4546.   74180.
12 USA         2  8768.   6580.   6227. 0.71  1848.   30158.
13 USA         3  5797.   4186    4098. 0.707 1175.   19030.
14 USA         4  4232.   3283.   2852. 0.674  831.   12994.
15 USA         5  2912.   2594.   1889. 0.648  543.    8729.
16 USA         6  1842.   1535.   1174. 0.637  325     5198.
17 USA         7   935.    809.    565. 0.604  185.    2560.
18 USA         8   488     375.    317. 0.649   88.1   1532
19 USA         9   260.    204.    178. 0.685   41.8    861.
20 USA        10   188     176.    103. 0.547   37.8    476.


Coefficient of Variation by age:

# A tibble: 10 x 3
     Age Russia_CV USA_CV
   <dbl>     <dbl>  <dbl>
 1     1     0.554  0.703
 2     2     0.547  0.71
 3     3     0.699  0.707
 4     4     0.654  0.674
 5     5     0.688  0.648
 6     6     0.728  0.637
 7     7     0.757  0.604
 8     8     0.917  0.649
 9     9     0.964  0.685
10    10     1.29   0.547
```

```
Age classes with correlation > 0.5: Age1, Age2, Age3, Age4, Age5, Age6

Age classes with correlation < 0.2: Age7, Age8, Age9, Age10

  Age1   Age2   Age3   Age4   Age5   Age6   Age7   Age8   Age9  Age10
 0.625  0.627  0.618  0.697  0.682  0.560  0.083  0.098  0.007 -0.500
```

4.2 Results comparing Russian and USA age-structured assessments

.

4.3 Comparison with the Gulf of Alaska (GOA) pollock estimates

Since the Russian context suggest some connectivity, for perspective, we thought it might be useful to conduct a similar comparison with results from the GOA pollock stock. Pollock assessments from the eastern Bering Sea (EBS) and Gulf of Alaska (GOA) show strong contrasts in both magnitude and dynamics of numbers-at-age. The EBS consistently supports far higher abundances across all ages and years (typically 5–15 times greater than the GOA; Figure 8). Recruitment pulses are evident in both regions, particularly at age 1, but the correlations by age class reveal only modest positive associations at younger ages, while older ages show little to no correspondence (Table 4.3). This indicates that early life stages may share some environmental drivers across basins, but overall population dynamics remain largely decoupled.

A more detailed breakdown by age class underscores these findings (Figure 9). Cohort strength in the EBS exhibits large variability with recurring strong year classes, while the GOA maintains lower overall abundance with weaker pulses. Time-series plots for ages 1–6 highlight this contrast, with the EBS showing sharp recruitment spikes and the GOA remaining more muted.

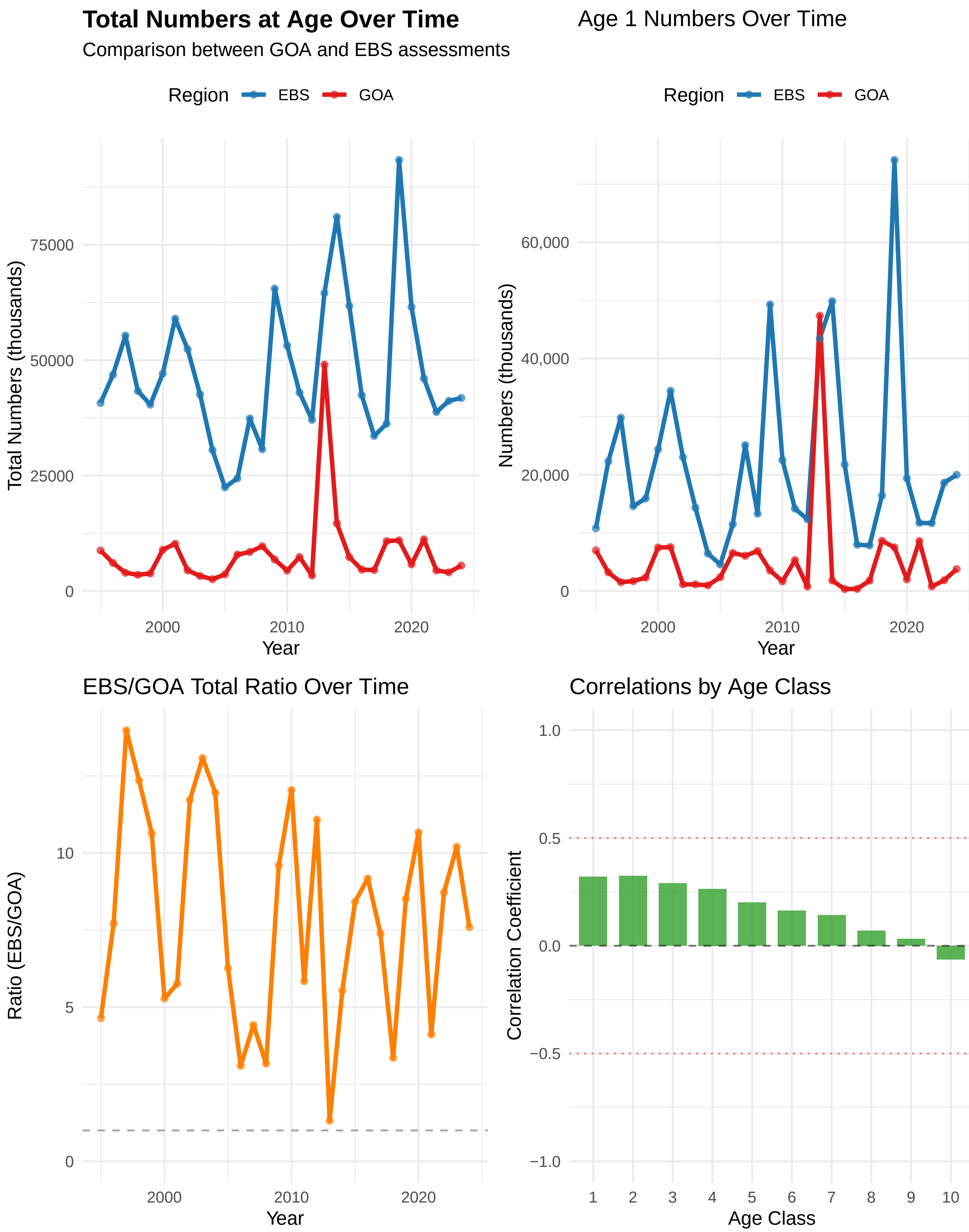
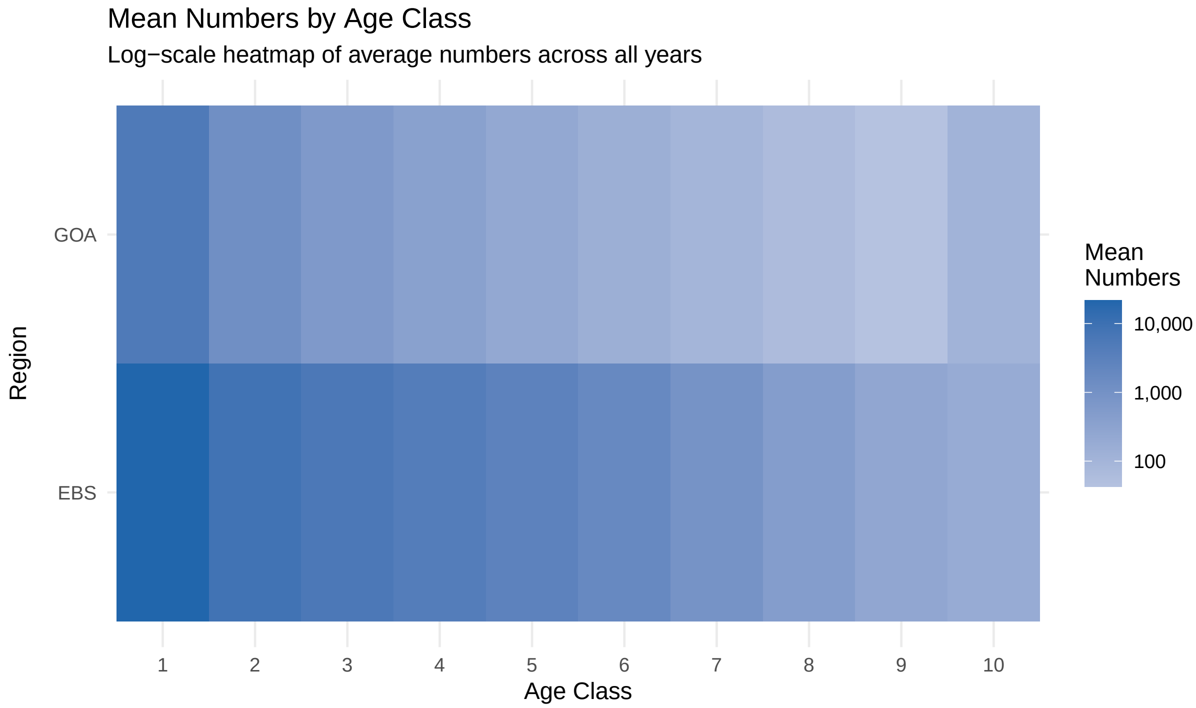


Figure 8: Various comparisons between GOA and EBS age-structured assessment results.



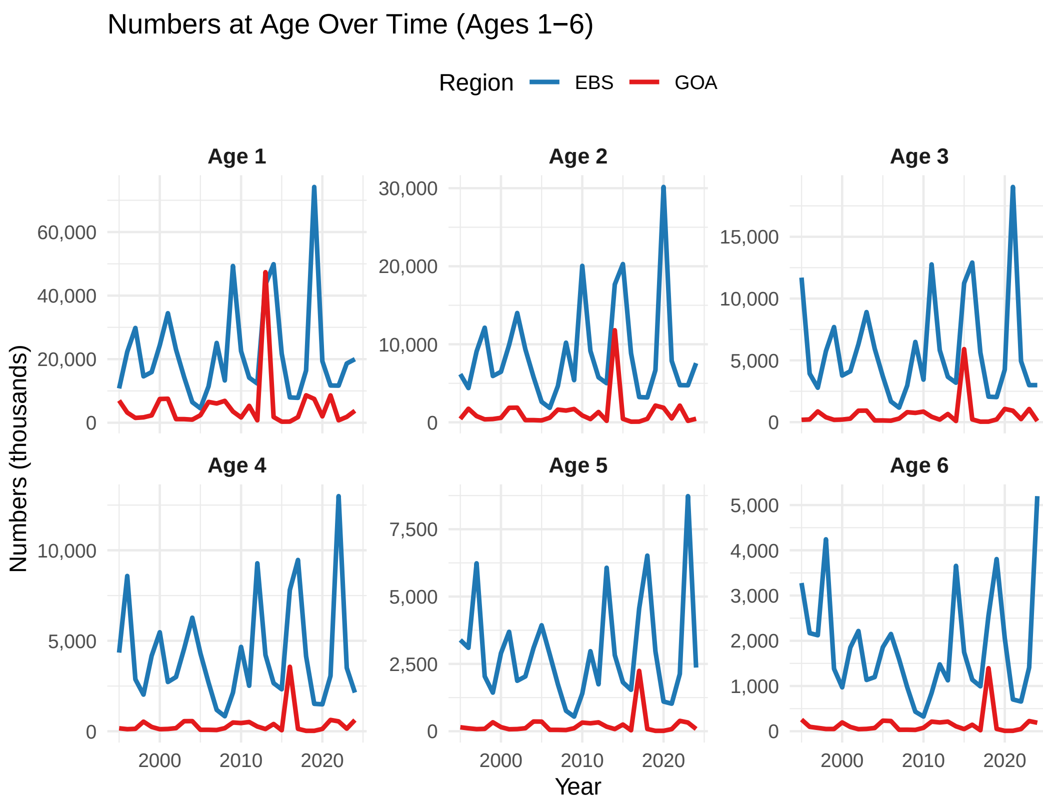


Figure 9: Various comparisons between GOA and EBS age-structured assessment results.

Table 4.3: Summary of age-specific correlations between GOA and EBS age classes, 1995-2024.

[

Correlations between GOA and EBS Age Classes] Correlations between GOA and EBS Age Classes

Age	Correlation
1	0.321
2	0.324
3	0.289
4	0.264
5	0.200
6	0.162
7	0.142
8	0.071
9	0.032
10	-0.064

5 Other items

The CIE noted in particular that some documentation of aspects of the ADMB code was incomplete. During the meetings, we began to more properly specify how the selectivity penalties were being calculated and implementated.

5.0.1 First-difference selectivity penalty (temporal)

We need the penalty used in ADMB where the **first year is compared to zero** (not to a previous estimated year). For a matrix of log-selectivities ($\log s_{y,a}$) (years($y = 1, \ldots, T$), ages ($a = 1, \ldots, A$), the penalty is:

$$\mathcal{P}_{\mathrm{FD,time}} = w \sum_{a=1}^{A} \Big[\big(\log s_{1,a} - 0 \big)^2 + \sum_{y=2}^{T} \big(\log s_{y,a} - \log s_{y-1,a} \big)^2 \Big].$$

RTMB/R implementation

```r
# log_sel: matrix [T x A] of log-selectivities (year rows, age cols)
# w: scalar or length-A vector of weights
fd_penalty_time <- function(log_sel, w = 1) {
  stopifnot(is.matrix(log_sel))
  T <- nrow(log_sel); A <- ncol(log_sel)
  if (length(w) == 1) w <- rep(w, A)
  stopifnot(length(w) == A)

  diffs <- rbind(log_sel[1, , drop = FALSE] - 0,   # year 1 vs 0
                 apply(log_sel, 2, diff))          # y>=2 diffs
  sum(colSums((sqrt(w) * diffs)^2))
}
```

5.0.2 Sanity check

```
set.seed(42)
T <- 5; A <- 3
L <- matrix(rnorm(T*A, sd=0.2), T, A); w <- c(1, 2, 0.5)

fd_loop <- function(L, w){
  acc <- 0
  for (a in seq_len(ncol(L))) {
    acc <- acc + w[a]*(L[1,a]-0)^2
    for (y in 2:nrow(L)) acc <- acc + w[a]*(L[y,a]-L[y-1,a])^2
  }
  acc
}

all.equal(fd_penalty_time(L, w), fd_loop(L, w))
```

```
[1] "Mean relative difference: 0.01953004"
```

5.0.3 First-difference selectivity penalty (age)

For within-year smoothing over age with a **zero anchor at the minimum age**:

$$\mathcal{P}_{\mathrm{FD,age}} = w \sum_{y=1}^{T} \Big[\big(\log s_{y,a_{\min}} - 0 \big)^2 + \sum_{a=a_{\min}+1}^{A} \big(\log s_{y,a} - \log s_{y,a-1} \big)^2 \Big].$$

RTMB/R implementation

```
# log_sel: matrix [T x A] of log-selectivities (year rows, age cols)
# a_min: column index of minimum age within 'log_sel' (default = 1)
fd_penalty_age <- function(log_sel, w = 1, a_min = 1) {
  stopifnot(is.matrix(log_sel))
  T <- nrow(log_sel); A <- ncol(log_sel)
  stopifnot(a_min >= 1, a_min <= A)
  if (length(w) == 1) w <- rep(w, T)
  stopifnot(length(w) == T)

  # Build diffs across age with anchor at a_min (assumes a_min=1; if not, set zeros below a_m
```

```
  diffs <- matrix(0, nrow = T, ncol = A)
  diffs[, a_min] <- log_sel[, a_min] - 0
  if (a_min < A) {
    for (a in (a_min+1):A) {
      diffs[, a] <- log_sel[, a] - log_sel[, a-1]
    }
  }
  sum(rowSums((sqrt(w) * diffs)^2))
}
```

5.1 Reference: survey lognormal scaling (consistency check)

When porting lognormal survey likelihoods, confirm whether ADMB's objective includes the normalizing constant and whether ( ) is on the **log** scale:

$\ell(\sigma) = -\sum_i \frac{(\log O_i - \log E_i)^2}{2\sigma^2} \; - \; n \log \sigma \; - \; \frac{n}{2} \log(2\pi).$

```
lognorm_nll <- function(obs, exp, sigma_log, include_const = TRUE) {
  r <- log(pmax(obs, .Machine$double.eps)) - log(pmax(exp, .Machine$double.eps))
  n <- length(r)
  nll <- sum(r^2) / (2 * sigma_log^2)
  if (include_const) nll <- nll + n*log(sigma_log) + n*0.5*log(2*pi)
  nll
}
```

References

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